

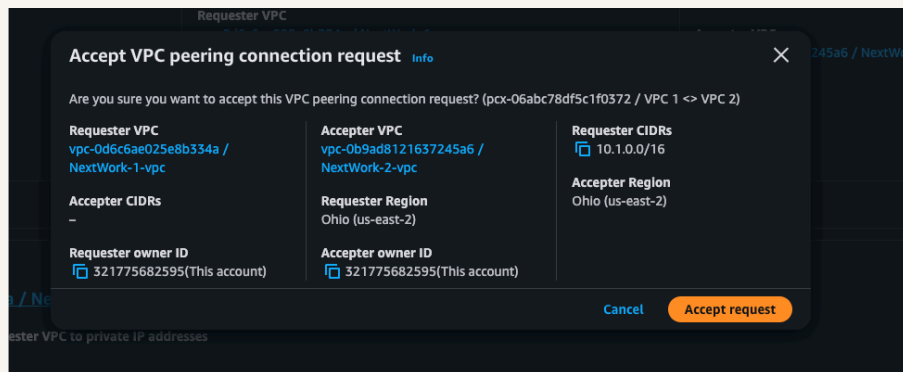


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VPC Peering



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Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a space in the amazon cloudspace allocated to us like a home in a city. it's useful because we have a fixed location in the cloudspace and are able to send and receive traffic data.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to setup a peering connection between two VPCs. Launched an EC2 instance in each VPC connection and was able to start a ping message between them.

One thing I didn't expect in this project was...

I didn't expect all the fun connection errors.

This project took me...

This project took me about 90 minutes.



In the first part of my project...

Step 1 - Set up my VPC

In this step, we create two VPCs. 1 IG for each VPC. 1 Public subnet each, 1 Network ACL and 1 Route table each.

Step 2 - Create a Peering Connection

In this step, we are setting up a connection between VPCs 1 and 2 by setting up a Peering connection.

Step 3 - Update Route Tables

In this step, we are setting up a route for traffic coming from VPC 1 to get to VPC 2 and vice versa.

Step 4 - Launch EC2 Instances

In this step, I am going to launch an EC2 instance in each of the VPCs 1 and 2.



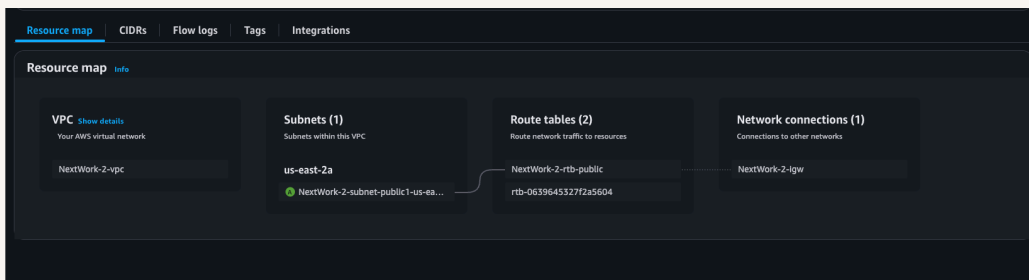
Multi-VPC Architecture

I started my project by launching two VPCs, both in one Availability Zone. I created two subnets - one for each VPC.

The CIDR blocks for VPCs 1 and 2 are 10.1.0.0/16 and 10.2.0.0/16. They have to be unique because having overlapping IP addresses could cause routing conflicts and connectivity issues.

I also launched 2 EC2 instances

I didn't set up key pairs for these EC2 instances as we will be using Amazon EC2 connect which manager the key pair for us.



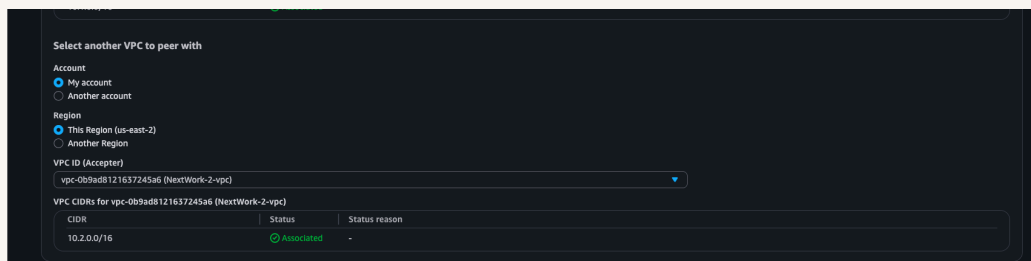


VPC Peering

A VPC peering connection is a direct connection between two VPCs.

VPCs would use peering connections to route traffic using their private IP addresses without using the public internet.

The difference between a Requester and an Acceptor in a peering connection is the one who initiates the peering connection and the one who accepts and establishes the peering connection.





Updating route tables

After accepting a peering connection, my VPCs' route tables need to be updated because we need a route for traffic in VPC 1 to know how to get to VPC 2 via the Peering connection.

My VPCs' new routes have a destination of 10.2.0.0/16 and 10.1.0.0/16. The routes' target was the Peering connection VPC 1 <> VPC 2.

Updated routes for rtb-0f7fb6beb9e0feb72 / NextWork-2-rtb-public successfully

rtb-0f7fb6beb9e0feb72 / NextWork-2-rtb-public

Details

Route table ID
rtb-0f7fb6beb9e0feb72

VPC
vpc-0b9ad8121637245a6 / NextWork-2-vpc

Main

No

Owner ID
321775682595

Explicit subnet associations

subnet-0495053ea16dc07cb / NextWork-2-subnet-public1-us-east-2a

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (3)

Filter routes

Both

Edit routes

Destination	Target	Status	Propagated
0.0.0.0/0	lgw-0622940b49e897fed	Active	No
10.1.0.0/16	pcc-05abc78d5c1f0372	Active	No
10.2.0.0/16	local	Active	No



In the second part of my project...

Step 5 - Use EC2 Instance Connect

In this step, we are connecting to the EC2 instance in VPC 1 using EC2 Instance Connect.

Step 6 - Connect to EC2 Instance 1

In this step, we are about to connect to the EC2 instance which has the public Elastic IP using EC2 Instance Connect.

Step 7 - Test VPC Peering

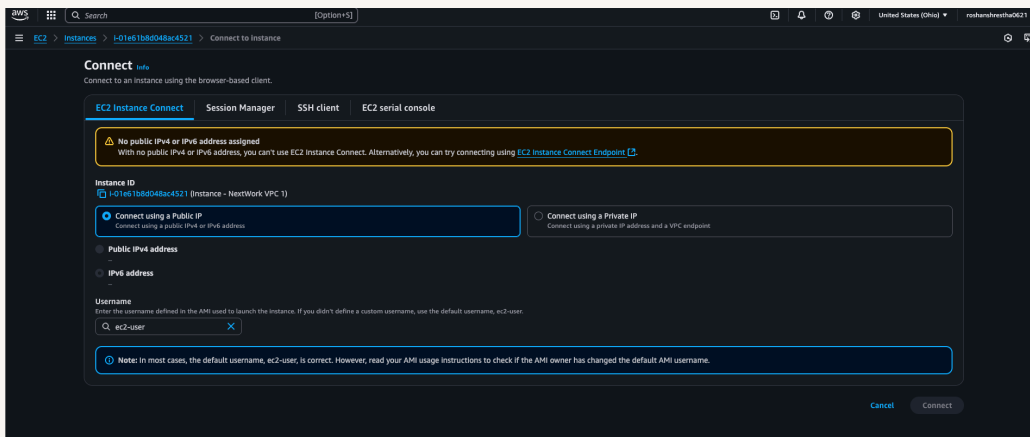
In this step, we use EC2 instance 1 to send messages to Instance 2 through the peering connection and solve any connection errors until instance 2 is able to respond back to instance 1.



Troubleshooting Instance Connect

Next, I used EC2 Instance Connect to connect to my EC2 instance on VPC 1 so that we can test peering connection with the EC2 instance on VPC 2.

I was stopped from using EC2 Instance Connect as I hadn't setup a public IP address for my instance in the public subnet. EC2 Instance Connect needs the public IP as it connects over the internet by default.

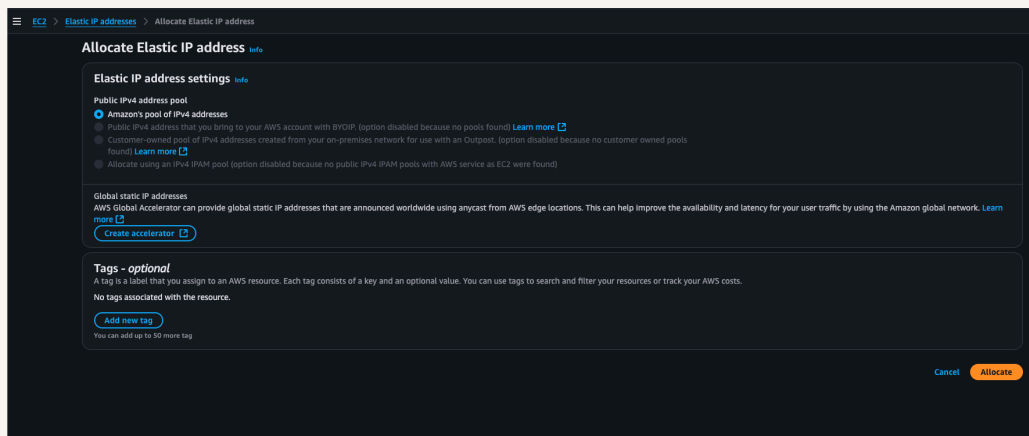


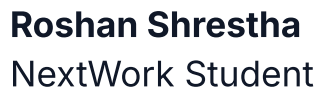


Elastic IP addresses

To resolve this error, I set up Elastic IP addresses. Elastic IP addresses are static IPv4 addresses that get allocated to your AWS account, and is yours to delegate to an EC2 instance.

Associating an Elastic IP address resolved the error because the EC2 instance on the public subnet of VPC 1 now has a public IP address. We can now use EC2 Instance Connect to connect to this EC2 instance.





A successful ping test would validate my VPC peering connection because my message from instance 1 is able to reach instance 2 through the peering connection and instance 2 is able to respond back using the same connection.

I had to update my second EC2 instance's security group because it needed to be able to accept the ICMP ipv4 message (ping message) from instance 1. I added a new rule that allows all ICMP IPv4 traffic from the public subnet of VPC 1, 10.1.0.0/16.

[illegible]



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